MIL-E-1/849F
20 February 1976
SUPERSEDING
MIL-E-1/849E
19 May 1970
(See note 32)

# MILITARY SPECIFICATION SHEET

## ELECTRON TUBE, MAGNETRON

## TYPE 6543

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, tunable frequency range 8,500 to 9,600 MHz, rated peak power output 65 kw, integral magnet

ABSOLUTE RATINGS:

| Parameter: | Ef | if (surge) | tk | tpc | ib | pi | Pi | VSWR | Tuner drive torque | T (anode) | T (input bushing) | Alt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | a | sec | μs | a | kw | W | --- | in.oz | °C | °C | ft |
| Maximum: | 13.7 | 10 | --- | 3.3 | 16.0 | 250 | 250 | 1.5:1 | 50 | 150 | 175 | 10,000 |
| Minimum: | --- | --- | 120 | --- | 14.0 | --- | --- | --- | --- | -65 | -65 | --- |

PHYSICAL CHARACTERISTICS:

| | | | |
|---|---|---|---|
| Dimensions: | See figure 1 | Cathode: | Unipotential |
| Mounting position: | Any | Magnet: | See note 4 |
| Mounting support: | Mounting flange | Weight: | 6 pounds 2 ounces (approx) |
| Marking: | See note 24 | Handling: | See note 4 |

TEST CONDITIONS:

| Parameter: | Ef | tpc | rrv | Du | Ib | VSWR |
|---|---|---|---|---|---|---|
| Unit: | V | μs | kv/μs | --- | mAdc | --- |
| Test condition 1: | | | | | | |
| Maximum: | --- | 0.40 | --- | --- | --- | 1.10 |
| | 10.4 | --- | --- | 0.0003 | 4.5 | --- |
| Minimum: | --- | 0.35 | 160 | --- | --- | --- |
| Test condition 2: | | | | | | |
| Maximum: | --- | 3.30 | --- | --- | --- | 1.10 |
| | 7.8 | --- | --- | 0.001 | 15.0 | --- |
| Minimum: | --- | 3.10 | 120 (Note 22) | --- | --- | --- |

| Frequency |
|---|
| F1 = 8,500 MHz |
| F2 = 8,600 MHz |
| F3 = 9,000 MHz |
| F4 = 9,375 MHz |
| F5 = 9,600 MHz |

GENERAL:

Qualification - Required



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| METHOD | REQUIREMENT OR TEST | NOTES | TESTS | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MINIMUM | MAXIMUM | |
| | Qualification inspection | | | | | | | |
| 1031 | Variable-frequency vibration | 6, 7, 8, 30, v | - | | --- | --- | --- | --- |
| 1042 | Shock, specified pulse | 6 | - | No voltages applied; 15G: t = 11 ms | --- | --- | --- | --- |
| 4223 | Tuner stop endurance | 9 | - | No voltages applied; | Torque | 50 | --- | in. -oz |
| 4223 | Mechanical tuning fatique | 7, 10, 11, 12, 31 | - | No voltages applied | --- | 10, 000 | --- | Cycles |
| 4223 | Operating torque or force | 10, c | - | TA = 25°C | Torque | --- | 15 | in. oz |
| 4223 | Resettability | 13 | 2 | F = 9, 550±50 MHz | ΔF | --- | 15 | MHz |
| 4266 | Direct-interelectrode capacitance | - | - | | C | 10 | 14 | pF |
| 4223 | Tuning rate | 14, c | 1 | T(anode) = 70° ± 10°C | ΔF | --- | ±60 | MHz |
| 4308 | RF bandwidth | 15, 16 | 1 | F = F1 to F5; Ib = 4.0 and 5.0 mAdc | BW | --- | 2.0/tpc | MHz |
| | | 21 | 2 | F = F1, F3, F5; Ib = 13.5 and 16.5 mAdc | BW | --- | 3.0/tpc | MHz |
| 4308 | Minor lobe ratio | 15, 16 | 1 | F = F1 to F5; Ib = 4.0 and 5.0 mAdc | Ratio | 8 | --- | dB |
| 4223 | Mechanical tuning rate | 17 | - | | --- | --- | --- | --- |
| 4028 | Barometric pressure, reduced | 18 | 2 | VSWR = 1.5:1; pressure = 500 mmHg (absolute) | --- | --- | --- | --- |
| 4027 | Temperature coefficient | c | 2 | T(anode) = 70° to 100°C; F = F3 | ΔF/ΔT | --- | 0.25 | MHz/°C |
| 1047 | Low-temperature operation | 19 | 2 | tk = 120 sec (max); F = F3 | MP | --- | 2.0 | % |
| 1143 | Forced cooling | c, 20 | 1 | | T | --- | 50 | °C(rise) |
| | Quality conformance inspection, part 1 | 1 | - | | | | | |
| 4003 | Pressurizing | - | - | 40 to 45 psia, input and output assemblies | --- | --- | --- | --- |
| 4289 | Heater current | - | - | Ef = 12.6 V; tk = 120 sec (max) | If | 2.0 | 2.4 | A |



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| METHOD | REQUIREMENT OR TEST | NOTES | TESTS | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MINIMUM | MAXIMUM | |
| | Quality conformance inspection, part 1 - Continued | | | | | | | |
| 4223 | Operating torque or force | 10 | - | TA = 25°±5°C | Torque | --- | 15 | in.-oz |
| 4223 | Mechanical tuning range | - | 1 | T(anode) = 70°±10°C | | | | |
| | | | | Upper limit | F | 9,600 | --- | MHz |
| | | | | Lower limit | F | --- | 8,500 | MHz |
| 4308 | RF bandwidth | 15, 16 | 1 | F= F1 to F5 | BW | --- | 2.0/tpc | MHz |
| | | 21 | 2 | F= F1 to F5 | BW | --- | 3.0/tpc | MHz |
| 4308 | Minor lobe ratio | - | 1 | | Ratio | 8 | --- | dB |
| 4315 | Stability | 15, 23 | 1, 2 | F = F1 to F5 | MP | --- | 1.0 | % |
| 4315 | Starting stability | 5, 21, 23 | 2 | F = F1 | MP | --- | 1.0 | % |
| 4306 | Pulse voltage | - | 2 | F = F5 | epy | 13.5 | 16.5 | kv |
| 4250 | Power output | - | 2 | t = 300 (max); F = F1, F3, F5 | Po | 65 | --- | W |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1031 | Variable-frequency vibration | 6, 2, v | - | No voltages applied | --- | --- | --- | --- |
| 4223 | Operating torque or force | 10 | - | TA = -50°C | Torque | --- | 25 | in.-oz |
| | | | | TA = +125°C; no voltages applied | Torque | --- | 15 | in.-oz |
| 4223 | Resettability | 13 | 1 | F= 9,550±50 MHz | ΔF | --- | 10 | MHz |
| 4310 | Frequency pulling figure | - | 1 | F = F1 to F5, in 100 MHz increments | ΔF | --- | 15 | MHz |
| 4250 | Power output | 16 | 2 | F = F1 to F5 | Po | 65 | --- | W |
| --- | Heater-voltage sensitivity | 25 | 1, 2 | | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | |
| --- | Life test | 3, 26, 27 | - | Group C; VSWR = 1.5:1(min) | --- | 1,334 | --- | Cycles |
| | | | | | t | 400 | --- | hrs |
| --- | Life-tests end points: | | | | | | | |
| 4250 | Power output | 28 | 2 | | Po | 50 | --- | W |
| 4308 | RF bandwidth | 28 | 2 | | BW | --- | 3.0/tpc | MHz |
| 4308 | Minor lobe ratio | 28 | 1 | | Ratio | 6 | --- | dB |
| 4315 | Stability | 28 | 1, 2 | | MP | --- | 2.0 | % |
| 4223 | Resettability | 28, 29 | - | | ΔF | --- | 50 | MHz |



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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

2. 10 to 55 to 10 Hz. Vibrate from 1 to 3 minutes at 10 G or 0.60-inch (1.53 mm) double amplitude; repeat for a total of 90 minutes in each of the three mutually perpendicular planes. Vibrate for 30 minutes at each of the resonant frequencies noted.

3. VSWR cycled through $\lambda g$ in about 30 minutes; starting at 8,500 MHz, the frequency shall be increased in 100 MHz increments every 10 hours.

4. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization.

5. Immediately after a nonoperating period of 168 hours minimum, the starting stability test shall be performed on 100 percent of the submitted lot.

6. After subjection to the specified test, the tube shall meet the requirements for rf bandwidth, minor lobe ratio, and stability as specified under test condition 1.

7. Separate samples shall be required for this test.

8. The tubes shall be fastened rigidly to the vibration platform and vibrated with simple harmonic motion. The direction of vibration shall be as indicated on figure 1.

9. The tuning mechanism shall withstand the specified torque at the limits of mechanical motion.

10. The torque required to operate the tuning shaft through the specified tuning range shall not exceed the maximum specified.

11. The tuner shall operate for at least the specified number of cycles. The tube shall then meet all of the conditions of test condition 2 and the torque and backlash at room temperature shall not exceed the specified amount. One cycle is defined as a complete traverse from one end of the tuning range to the other and back again. During this test the tuner may be lubricated not more than once every 1,000 cycles with Texaco low-temperature grease No. 1,888, or equal.

12. After the specified number of cycles, the tube shall pass all the tests under test condition 2 in quality conformance inspection, part 1.

13. The frequency obtained by turning the tuner to a given setting in one direction shall be reproducible within specified limits when returning to that same setting from the opposite direction after thermal equilibrium.

14. After thermal frequency stability has been reached at each of the frequencies of this test, F1, F2, F3, F4, F5 (see note 24) the curve of frequency versus tuner position shall not deviate more than the specified amount from the curve on figure 2.

15. This test shall be performed by a continuous observation while the tube frequency and a stub having a VSWR of 1.5:1 minimum are continuously varying. The stub shall travel a minimum of 180 electrical degrees for every 20 MHz of magnetron tuning.

16. The measurement shall be made throughout the specified tuning range at a rate sufficiently slow to permit steady-state indications. At no place in the specified tuning range shall the specified limits be exceeded.



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NOTES: - Continued

17. With the tuning dial adjusted for a frequency of 9,000 MHz, an operating frequency of 9,600 MHz shall be obtained by clockwise rotation of the designated end of tuning shaft 68 ± 12 turns. A frequency of 8,500 MHz shall be attained by turning the shaft 49 ± 6 turns in the opposite direction from the 9,000 MHz setting.

18. With the specified input pressurization, or output pressurization, or both, there shall be no evidence of breakdown in the input assembly, or output assembly.

19. The missing pulses (MP), due to any causes, are considered to be missing if the rf energy is less than 70 percent of the normal energy level in the frequency range of 8,900 to 9,100 MHz. The VSWR shall be adjusted to that phase producing maximum instability and the MP counted during the last 3 minutes of a 5-minute test period.

20. An airflow of 10 cubic feet per minute (cfm) at approximately 760 mmHg shall be directed on the cooling fins from an orifice 2-1/2 inches by 1-3/16 inches (63.50 by 30.16 mm). Temperature of the anode shall be measured and the rise shall not exceed the amount specified.

21. The tube shall be operated into a transmission line with a VSWR of 1.5:1 minimum adjusted in phase to produce the maximum spectrum degradation.

22. The rrv shall be expressed in kv per μs defined by the steepest tangent to the leading edge of the voltage pulse above 80 percent amplitude. Any capacitance used in viewing system shall not exceed 6.0 pF. No spike or ripple shall exceed ± 7 percent of the average peak value of voltage or current. Inverse voltage shall not exceed 20 percent of the forward voltage. The negative amplitude of any post pulse voltage oscillation shall not exceed 10 percent of the average peak value.

23. The VSWR shall be adjusted to the phase producing maximum instability and the MP counted for a period of 3 minutes starting 30 seconds after application of high voltage.

24. In addition to regular markings, letters and dial settings shall be stamped on the tube at the point specified on figure 1, to indicate F1 through F5 ± 10 MHz at the start of life under test condition 2 conditions with the body temperature 70° ±10°C.

25. All of the requirements of test conditions 1 and 2 shall be met with the heater voltage at ±8 percent from the voltage specified for test.

26. Life test shall be run with heater power frequency set at the major resonant frequency, if any, as determined by the heater structure type test, provided the resonant frequency occurs at 55 to 65 Hz or in the interval 340 to 1,800 Hz. If there is no such resonant frequency, heater power may be at any convenient frequency up to 1,800 Hz at the discretion of the manufacturer. To measure the resonant frequency, one of the two following methods shall be used:

    (a) Supply the specified preheating voltage to the tube heater from a variable frequency power supply. Vary the supply frequency while listening (with a sensitive stethoscope for any indication of resonance. At the resonant frequency distinct vibration of the structure will be heard.

    (b) Supply heater voltage to the tube from a variable frequency supply. Operate the tube (test conditions 1) and observe the rf spectrum while varying the frequency of the heater voltage. At the resonant frequency of the heater cathode structure, large values of frequency modulation will be observed.

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NOTES: - Continued

27. Life test shall be conducted in accordance with the following cycle:

| Test condition | Ef (V) | Duration |
|---|---|---|
| Standby | 12.6 | 2 minutes |
| 1 | 10.4 | 3 minutes |
| 2 | 7.8 | 15 minutes |
| Off | --- | 16 minutes |

28. If during life test, the tube does not meet the specified limits it shall be recycled for an additional 5 cycles. At such time the tests shall be repeated. If the tube fails the second test, it shall be considered unsatisfactory.

29. At each of the dial settings called for in note 24, the measured frequency shall not differ from the stated frequency by more than the amount specified.

30. 5 G or DA = 0.01 inch, whichever is limiting value. Vibration frequency = 5 to 500 to 5 Hz varied uniformly over complete range in 15 minutes, repeat for a total of 2 hours in each of three planes.

31. Duration of each cycle = 45 seconds (maximum).

32. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - WC, WE
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3010)



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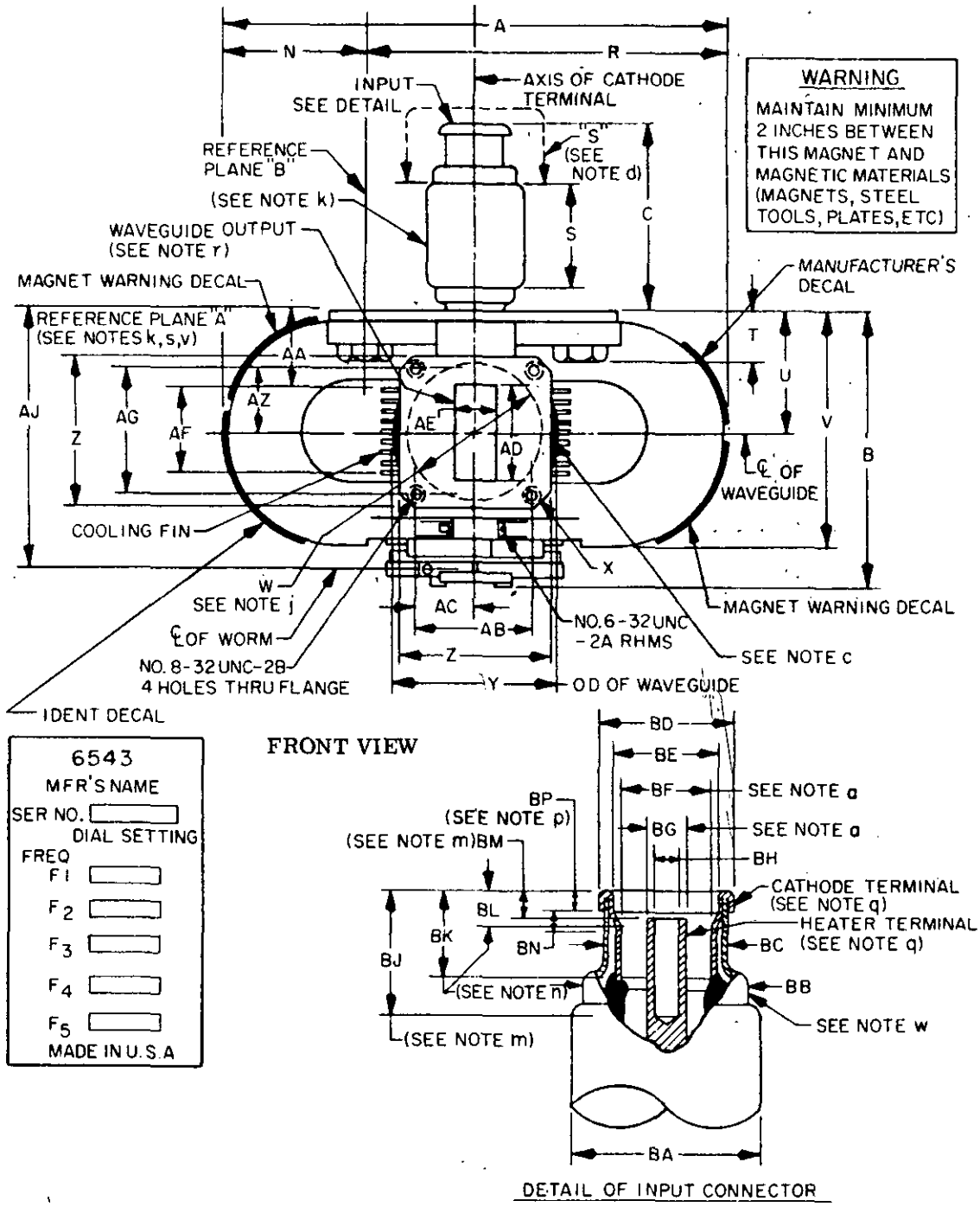



FIGURE 1. Outline drawing of electron tube type 6543.



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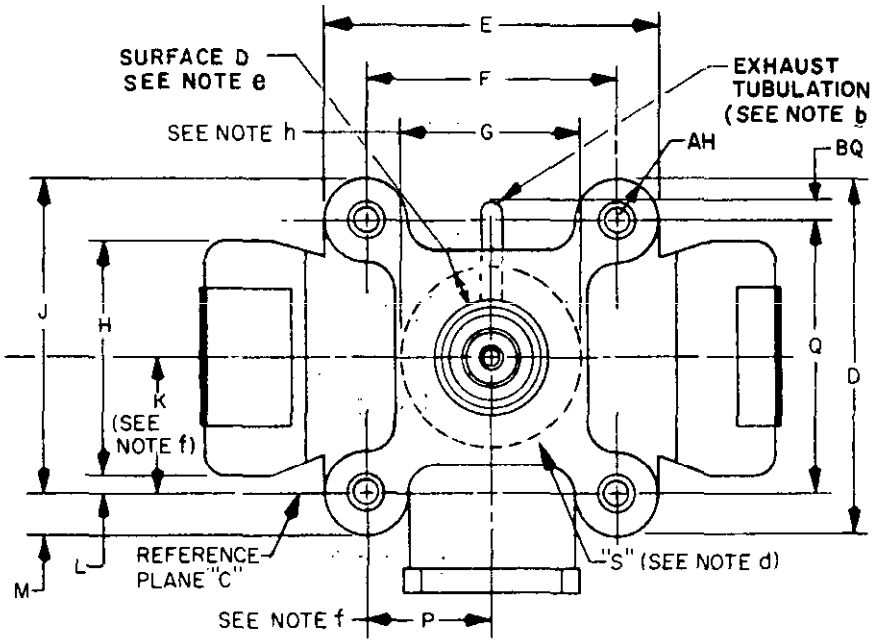
| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 1 (see note 1) | | |
| A | | 5.938 (150.83) |
| B | | 3.232 (82.09) |
| C | 2.094 (53.19) | 2.219 (56.36) |
| N | | 1.688 (42.88) |
| R | | 4.250 (107.95) |
| S | 1.250 (31.75) | |
| T | .594 (15.09) | .656 (16.66) |
| U | 1.386 (35.20) | 1.426 (36.22) |
| Y | | 1.906 (48.41) |
| AB | 1.470 (37.34) | 1.478 (37.54) |
| AC | .732 (18.59) | .742 (18.85) |
| AG | 1.348 (34.24) | 1.356 (34.44) |
| AJ | 2.984 (75.79) | 3.016 (76.61) |
| AZ | .671 (17.04) | .681 (17.30) |
| BA | | 1.188 (30.18) |
| BB | .995 (25.27) | 1.005 (25.53) |
| BC | .745 (18.92) | .755 (19.18) |
| BD | .825 (20.96) | .838 (21.29) |
| BE | .645 (16.38) | .655 (16.64) |
| BF | .532 (13.51) | .545 (13.84) |

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 1 (see note 1) | | |
| BG | .234 (5.94) | .266 (6.76) |
| BH | .164 (4.17) | .174 (4.42) |
| BJ | .750 (19.05) | |
| BK | .516 (13.11) | |
| BL | | .200 (5.08) |
| BM | .125 (3.18) | .187 (4.75) |
| BN | .109 (2.77) | .141 (3.58) |
| BP | .115 (2.92) | .135 (3.43) |
| **Reference dimensions** | | |
| V | 2.781 (70.64) | |
| W | 1.625 (41.28) DIA | |
| X | 1.156 (29.36) RAD | |
| Z | 1.830 (46.48) | |
| AA | .922 (23.42) | |
| AD | 1.122 (28.50) | |
| AE | .497 (12.62) | |
| AF | .984 (24.99) | |

FIGURE 1. Outline drawing of electron tube type 6543 - Continued.



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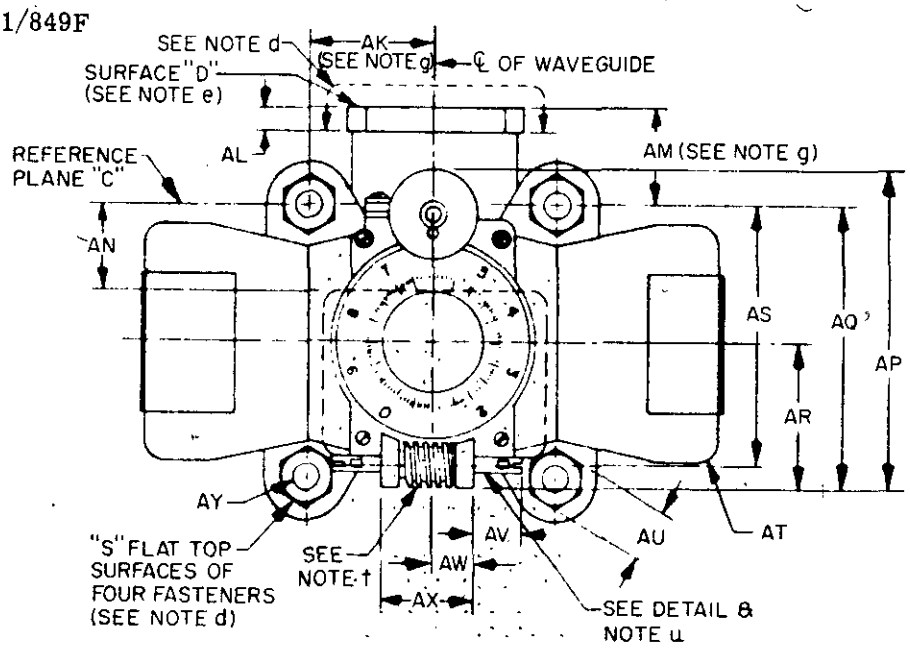

BOTTOM VIEW

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 1 (see note 1) | | |
| D | | 3.625(92.08) |
| E | | 3.438(87.33) |
| F | 2.521(64.03) | 2.541(64.54) |
| G | 1.875(47.63)DIA | |
| H | | 2.425(61.60) |
| J | | 3.203(81.36) |
| L | .178 (4.52) | .258 (6.55) |
| M | | .422(10.72) |
| Q | 2.771(70.38) | 2.791(70.89) |
| Quality conformance inspection, part 2 | | |
| BQ | | .125(3.18) |
| **Reference dimensions** | | |
| P | 1.265(32.13) | |
| AH | .375 (9.53) RAD | |
| K | 1.391(35.33) | |

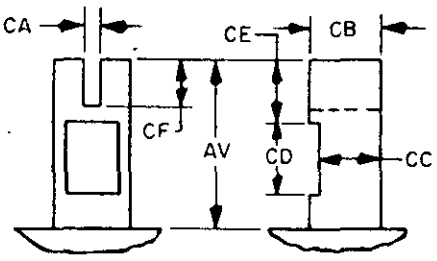
FIGURE 1. Outline drawing of electron tube type 6543 - Continued.



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TOP VIEW




DETAIL OF SHAFT END

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 1 (see note 1) | | |
| AK | 1.250 (31.75) | 1.280 (32.51) |
| AM | .990 (25.15) | 1.040 (26.42) |
| AP | 3.243 (82.37) | 3.293 (83.64) |
| AQ | 2.844 (72.24) | 2.938 (74.63) |
| AR | 1.490 (37.85) | 1.510 (38.35) |
| AS | 2.594 (65.89) | 2.688 (68.28) |
| AT | .094 (2.39) RAD | |
| AV | .428 (10.87) | .448 (11.38) |
| AX | 1.000 (25.40) | 1.016 (25.81) |
| AY | .276 (7.01) DIA | .286 (7.26) DIA |
| CA | .040 (1.02) | .045 (1.14) |
| CB | .187 (4.75) DIA | .190 (4.83) DIA |
| CC | .151 (3.84) | .161 (4.09) |
| CD | .188 (4.78) | .203 (5.16) |
| CE | .141 (3.58) | .156 (3.96) |
| CF | .115 (2.92) | .135 (3.43) |
| **Reference dimensions** | | |
| AL | .250 (6.35) FLANGE | |
| AN | .797 (20.24) | |
| AU | .500 (12.70) | |
| AW | .437 (11.10) | |

FIGURE 1. Outline drawing of electron tube type 6543 - Continued.



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NOTES:
a. Gold, silver, or nickel plate over full extent of maximum plug penetration.
b. Maximum projection of the tubulation shall not extend more than .125(3.18) beyond the centerline of the two mounting holes on the side opposite the waveguide flange.
c. Reference point for anode T measurements locate where a central fin meets the anode block near the output section.
d. All metal surfaces shall be painted except those designated "S" and "D". Surfaces designated "S" shall be gold, silver, nickel, or bright alloy plated, or brass surfaces except as specified.
e. Hermetic connections may be made to surface "D".
f. Axis of cathode terminal shall occupy specified location within .047 (1.19) RAD. Note g shall apply.
g. Limits include angular and lateral deviations.
h. With diameter "G" resting on a plane surface coincident with reference plane "A", a gage .010 (.25) thick and .125 (3.18) wide shall not enter, and areas of the base plate outside diameter "G" shall be within .010 (.25) of the plane surface.
j. With diameter "W" resting on a plane surface, a gage .005 (.13) thick and .125 (3.18) wide shall not enter.
k. Any portion of the assembly which extends below reference plane "A" shall be within .625 (15.88) RAD of specified axis of input.
m. Defines extremity of cylindrical section designated by .169 (4.29) diameter.
n. Defines extremity of cylindrical section designated by .540 (13.72) diameter.
p. Clamping shall be confined to this area.
q. Heater terminal and cathode terminal shall be concentric within .010 (.25).
r. Protective covers shall be provided with tube.
s. Reference planes "A", "B", and "C" are mutually perpendicular.
t. Gear and worm threads shall be free from corrosion, paint, and other imperfections. All tuner parts except worm, shaft, and striking pin shall be nickel plated.
u. A sleeve .195 (4.95) inside diameter by .406 (10.31) outside diameter by 1.00 (25.40) long must pass over ends of shaft to face of worm bracket.
v. For vibration tests, planes of vibration shall be reference planes "A", "B", and "C".
w. Reference point for cathode bushing temperature measurements. Locate on 1.000 (25.40) diameter cathode terminal.

FIGURE 1. Outline drawing of electron tube type 6543 - Continued.



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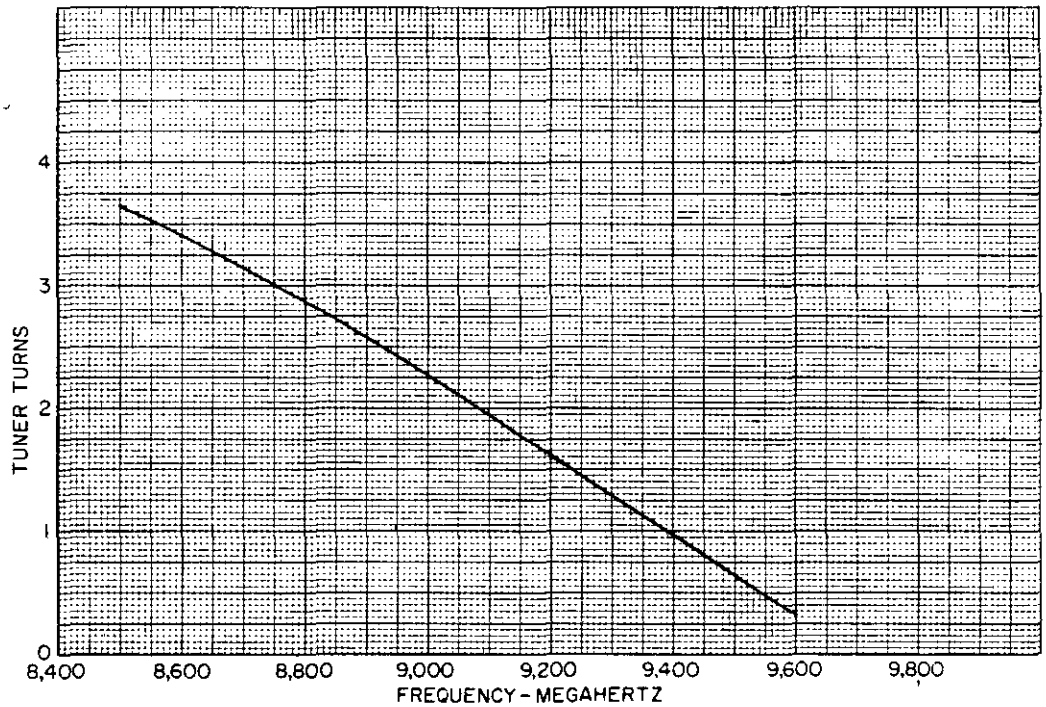



FIGURE 2. Tuning linearity.



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